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(54) Title (EN): HOCI-PFP-ALGINATE-CHITOSAN NANOPARTICLES PRODUCTION METHOD AS A LUNG CANCER ANTIBODY

(54) Title (FR): PROCÉDÉ DE PRODUCTION DE NANOPARTICULES D'HOCI-PFP-ALGINATE-CHITOSANE EN TANT QU'ANTICORPS CONTRE LE CANCER DU POUMON

(57) Abstract:

(EN): This invention is related to the fields of medical biotechnology, nanobiotechnology, and nanomedicine. It involves a nanomedicine product that can release HOC1 (hypochlorous acid) at the tumor site in a controlled manner to treat lung cancer. HOC1 is a potent antimicrobial and antitumor agent that can target tumor tissues while minimizing damage to healthy tissues. The produced nanoparticles have a two-layer structure: the inner layer contains alginate, PFP (perfluoropentane), and $Ca(OCl)_2$ to release HOC1, and the outer layer has a chitosan coating with folate and salicylic acid for targeting and proton release. The production process involves nanoemulsion and coating with the ionic gelation technique. When the nanoparticles are delivered to the lungs and targeted to cancer cells with folate, ultrasound radiation triggers the release of HOC1 to induce apoptosis and destroy cancer cells effectively, reducing side effects and improving treatment outcomes for lung cancer patients.

(FR): Cette invention se rapporte aux domaines de la biotechnologie médicale, de la nanobiotechnologie et de la nanomédecine. L'invention concerne un produit de nanomédecine qui peut libérer de l'HOCl (acide hypochloreux) au niveau du site tumoral de manière contrôlée pour traiter le cancer du poumon. L'HOCl est un agent antimicrobien et antitumoral puissant qui peut cibler les tissus tumoraux tout en réduisant au minimum les dommages aux tissus sains. Les nanoparticules produites présentent une structure à deux couches : la couche interne contient de l'alginate, du PFP (perfluoropentane) et du $Ca(OCl)_2$ pour libérer l'HOCl, et la couche externe présente un revêtement de chitosane avec du folate et de l'acide salicylique pour le ciblage et la libération de protons. Le procédé de production implique une nanoémulsion et un revêtement mettant en œuvre la technique de gélification ionique. Lorsque les nanoparticules sont administrées aux poumons et ciblées vers les cellules cancéreuses avec le folate, un rayonnement ultrasonore déclenche la libération d'HOCl pour induire l'apoptose et détruire efficacement les cellules cancéreuses, ce qui permet de réduire les effets secondaires et d'améliorer les résultats du traitement pour les patients atteints d'un cancer du poumon.

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